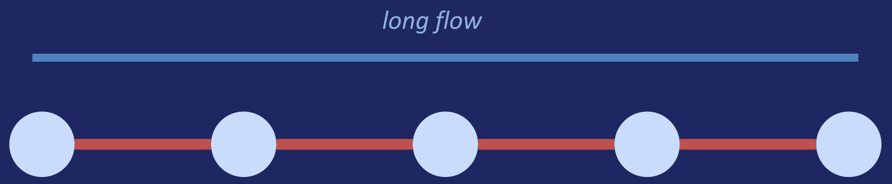

Multi-Bottleneck Fairness for High-Precision Congestion Control

HPCC-FS — one fair-rate field, one scalar per switch port

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BACKGROUND

HPCC in 60 seconds: precision congestion control

1. **Switch stamps raw per-hop state into INT**
   tx bytes, queue length, link rate — stateless per flow

2. **Sender computes exact utilization U**
   multiplicative update toward target η = 95%

3. **Near-zero queues, top-tier FCT**
   beats DCQCN / TIMELY / DCTCP on single-bottleneck benchmarks

the INT header

```
hop 1: tx, qlen, rate
hop 2: tx, qlen, rate
hop 3: tx, qlen, rate
… up to maxHop = 5
```

8 bytes per hop — the record grows with path length

The catch: the sender sees load, not its fair share.

MOTIVATION

Modern traffic crosses many contended links in series

Cross-pod ML
allreduce spans aggregation + core layers

Disaggregation
memory / storage traffic transits multiple fabrics

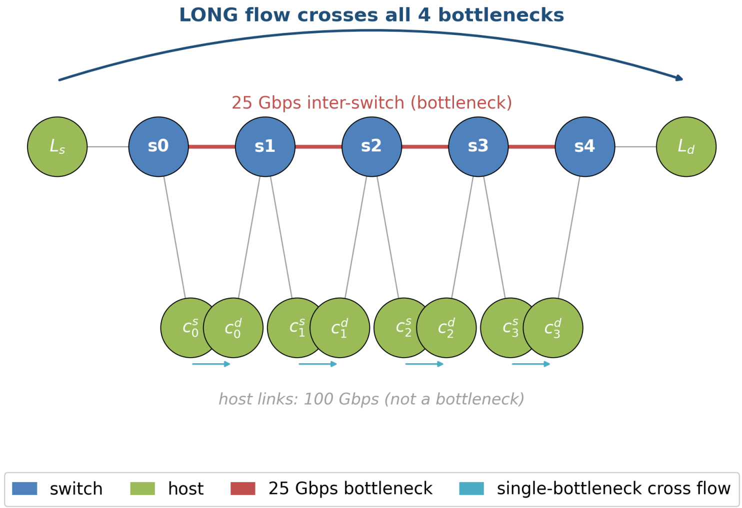
Multi-tenancy
shared links at every tier of the DC

Does HPCC still deliver fair service when a flow crosses multiple bottlenecks — not just one?

Nobody had measured it carefully. We did — deterministically, against a max-min oracle.

The parking lot, scored against a max-min oracle

Parking lot at N=4: one LONG flow + one CROSS flow per bottleneck



Setup
N bottlenecks, 25 Gbps each. One long flow crosses all N; one cross flow per link.

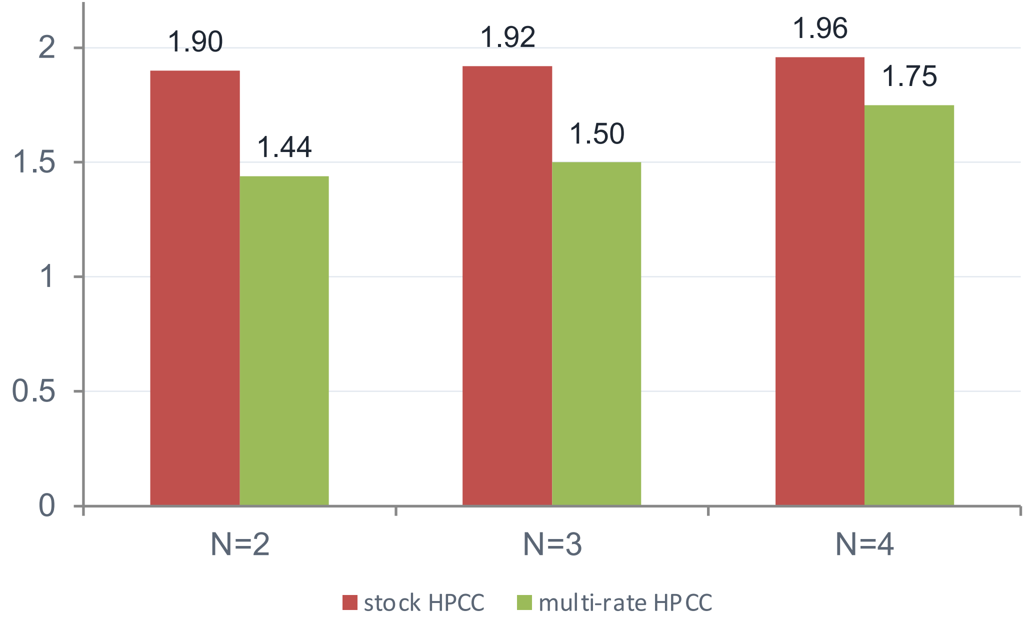
Fair answer
Every flow gets 12.5 Gbps (progressive filling, recomputed as flows finish).

Metric
unfairness = long FCT / mean short FCT. Oracle = 1.0

Deterministic: every configuration reproduces byte-identical artifacts.

FINDING 1

The long flow is starved ≈ 2× — and it's structural



unfairness ratio — max-min oracle = 1.0

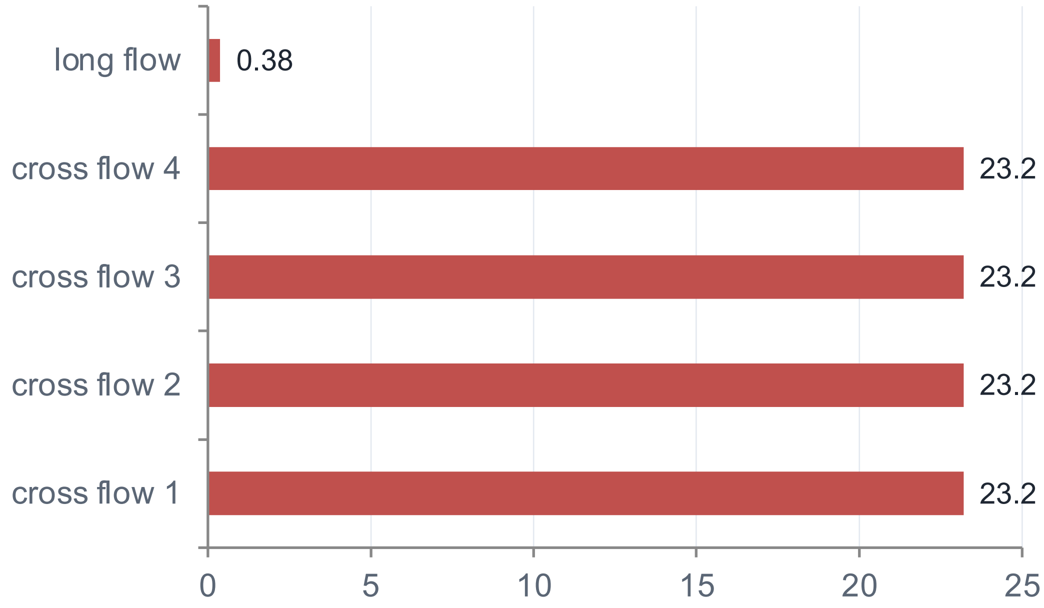
1.96× penalty at N = 4

3.5× when the long flow is small (RPC-scale)

N ≥ 5 breaks outright: INT maxHop overflow

FINDING 2

Winner-take-all — and nothing is ‘wrong’



stock HPCC, N = 4, first 18 ms — fair share is 12.5 Gbps for everyone

Zero PFC. Zero loss.

No pathology to point at — every link is at target utilization η.

The unfair allocation is the control law's equilibrium.

At each shared link the local single-bottleneck flow wins; the long flow is squeezed at all of them.

NEGATIVE RESULT

Five sender-only fixes — none gets close

sender-side variant (mb_mode)	idea	best penalty @ N=4
k-aware additive increase	scale AI by congested-hop count	1.63×
debiased max/mean blend	soften the max-hop bias	1.95×
responsibility-weighted MD	MD scaled by own share	1.89×
k-th-root MD	split the decrease across hops	1.92×
global share-aware AI	AI boosted by (1 − share)	1.71×

Each variant is a 1–2 line change in HPCC's update *(mb_mode 0 stays byte-identical to stock)* — the target is 1.0×.

WHY

Two structural reasons sender-only cannot work

1 · The η-hold equilibrium

Links settle at target utilization → HPCC's multiplicative term ≈ 1.

Every flow holds whatever rate it grabbed at startup. Additive nudges need ~10^4 RTTs to redistribute.

2 · N is not observable

Fair share is C / N. INT tells the sender C — never N, the link's flow count.

Without N there is no strong multiplicative move toward fair share — only slow AIMD nudging.

The information isn't at the sender — so the fix can't be either. It belongs at the link.

THE FIX, PART 1

RCP: the link advertises one fair rate to every flow

$$R \leftarrow R \cdot (1 + (\alpha \cdot (C - y) - \beta \cdot q/d) / C)$$

$\alpha \cdot (C - y)$

chases spare capacity — link under-full ⇒ R rises

$\beta \cdot q/d$

brakes on backlog — queue forms ⇒ R falls

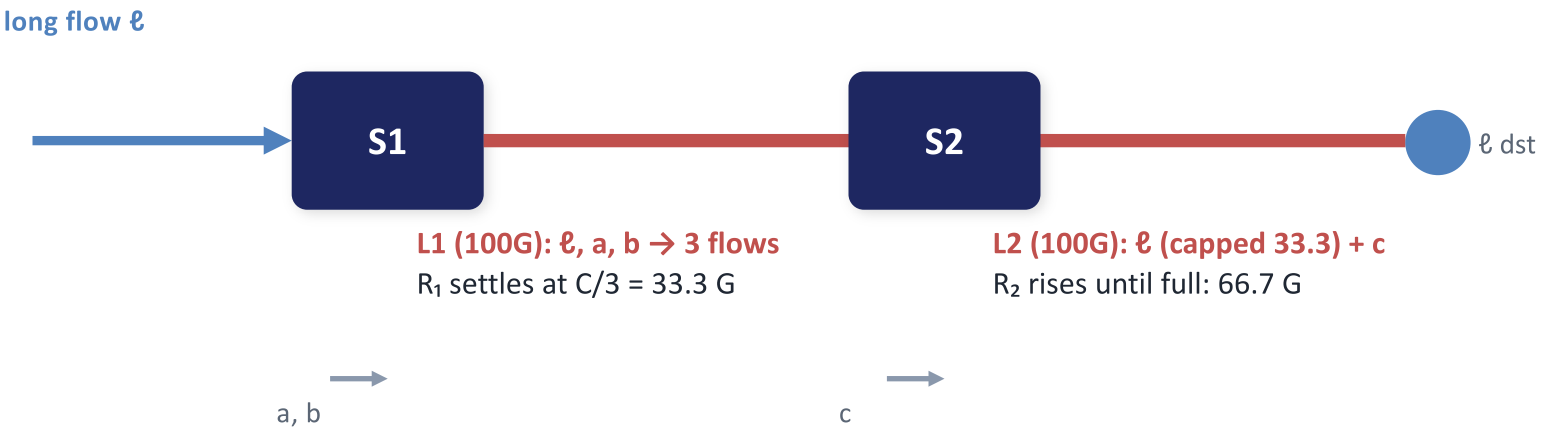
fixed point

$y = C$, $q = 0$ ⇒ $N \cdot R = C$ ⇒ $R = C/N$

The link converges to the max-min share *without ever counting flows* — N is implicit in the observed load y.

THE FIX, PART 2

Worked example: path-min gives network-wide max-min



ℓ pkt: fairRate 0 → **33.3** → min(33.3,66.7) = **33.3**

ℓ = a = b = 33.3

c = 66.7

Exactly the max-min allocation — both links 100% full, no per-flow state anywhere.

The whole mechanism: three small pieces

Switch

- one scalar fair rate per egress port
- RCP update once per control interval (≈ RTT)
- state: 3 numbers — no per-flow data

Wire

- new INT mode: one 64-bit fairRate field
- 8 B / packet, independent of hop count
- each hop min-aggregates; receiver echoes

Sender

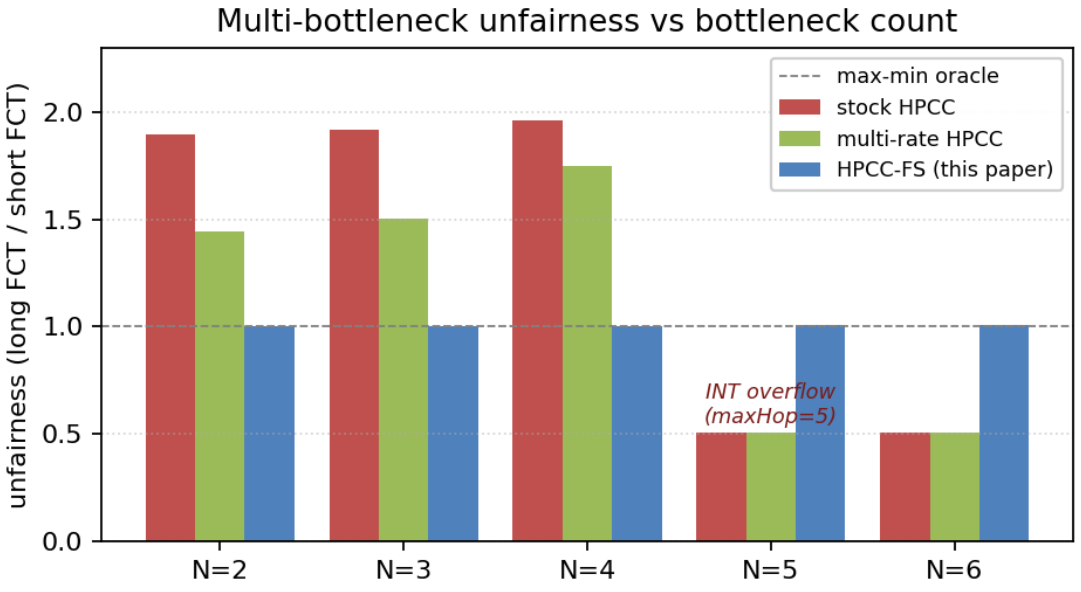
- new ACK handler: rate ← path-min fairRate
- clamped to [minRate, NIC]
- rate-only: per-flow window cap off

stock HPCC (cc_mode 3): byte-for-byte unchanged

one 8-byte field replaces the per-hop record

RESULTS

Flat at 1.0 across the whole sweep — zero PFC



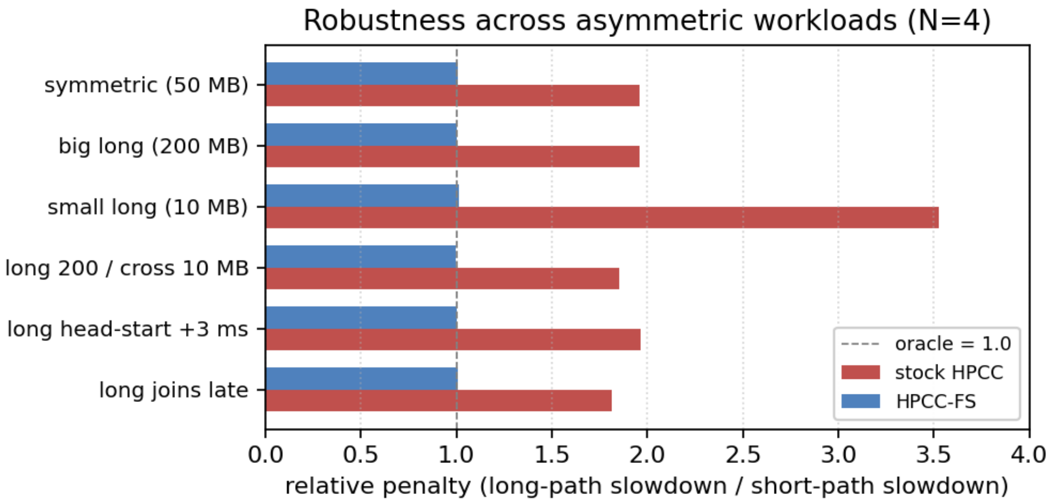
1.005×
penalty at N = 4 (was 1.96×)

1.003–1.007×
across N = 2…6, incl. the overflow regime

0 PFC
every workload, every N

HPCC-PINT (compressed INT) still shows 1.77–1.88× — a smaller footprint is not the fix; a fair-share signal is.

It generalizes — and holds under asymmetry



six asymmetric-size / staggered-start variants at N = 4: HPCC-FS within 1.0 ± 0.012 (worst stock case 3.53× → 1.012×)

topology	stock	HPCC-FS
3-tier tree fabric	1.36×	1.003×
k=4 ECMP fat-tree †	1.34×	1.001×
parking lot N=5,6	breaks	1.006–1.007×

† oracle-normalized (removes ECMP placement luck). Path-min is path-invariant — ECMP needs no design change.

IS IT A TRICK?

Ablation, sensitivity, and HPCC’s home turf

Rate-only is necessary

Re-enable the per-flow window cap → 1.5–2.5×, worse with path length.

var_win sizes the window from NIC line rate, not the path — the long flow gets window-bound below fair.

Parameters not load-bearing

RCP defaults α = 0.4, β = 0.226 — untuned.

Sweeping α, β, startup fraction, minRate one-at-a-time: penalty stays 1.004–1.008×, PFC = 0 throughout.

Home turf: not a trade-off

Single-bottleneck incast, 8 Gbps startup:

mean FCT 242 vs 259 µs
peak queue 38 vs 143 KB

beats HPCC’s FCT at ¼ the queue — fairness headline unchanged.

Everything regenerates deterministically from the artifact — figures and tables are built from live runs.

POSITIONING

What this is — and what it isn't

1. **The mechanism is RCP.** We claim the diagnosis and the placement, not a new control law. Placing one fair-rate signal at the switch is what closes the gap.

2. **Simulation-only, three topology families.** No testbed; no head-to-head with Bolt / PowerTCP (would require re-implementing them). The simulator re-port reproduces the original HPCC baselines end-to-end.

3. **Convergence is ~4 ms, not one RTT.** Fast and path-length-independent — the per-port estimate needs ~10^2 RTTs to settle.

Future: a hybrid overlay — `rate = min(rate_HPCC, fairRate)` — keep HPCC's precision and the fairness; needs a path-aware window.

CONCLUSION

One field, one scalar — fairness restored

Diagnosis	HPCC starves multi-bottleneck flows ~2× (3.5× small flows); breaks past 4 hops. Zero PFC — it's the equilibrium.
Negative result	Five sender-only fixes fail: at η-hold, no strong move exists without the flow count N.
HPCC-FS	One RCP fair-rate field, one scalar per port → 1.005×, zero PFC, robust, stock path untouched.

Artifact (code · paper · reproduction · data): `github.com/UNOCourseDemo/hpcc-fs`

Thank you — questions?

BACKUP

The re-ported simulator reproduces SIGCOMM ’19

short-flow p99 slowdown	HPCC	DCQCN	TIMELY	DCTCP	PINT
WebSearch 50% load	1.9	13.7	18.3	5.7	2.1
FB-Hadoop 70% load	6.1	30.6	**221.9**	8.4	5.0
PFC events (fb 70%)	**0**	0	**1.12 M**	0	0

5 schemes × 2 workloads × 3 loads on a 376-node fat-tree (320 servers, 100 G NICs).

All of the paper’s signatures reproduce: HPCC’s short-flow tail far below DCQCN/TIMELY, sub-MB queues, zero PFC; TIMELY’s PFC explosion at load; DCTCP intermediate. Along the way we found and fixed a latent uninitialized-member bug (Node::m_node_type) exposed only at scale.

BACKUP

Mixed-size workloads · the INT-overflow artifact

Mixed sizes (WebSearch dist.)

Long-flow mean slowdown −15%; short-path mean +45%.

This is correct max-min: stock's starvation of the long flow was inflating headroom that short flows enjoyed.

Per-class targets (beyond pure max-min) are application policy — future work.

Why stock 'improves' at N ≥ 5

The INT header holds maxHop = 5 per-hop records. Longer paths overflow the record; the utilization estimate is computed over the wrong hops.

The 0.51× 'advantage' is corrupted control input, not fairness. HPCC-FS is immune: one field, any path length.